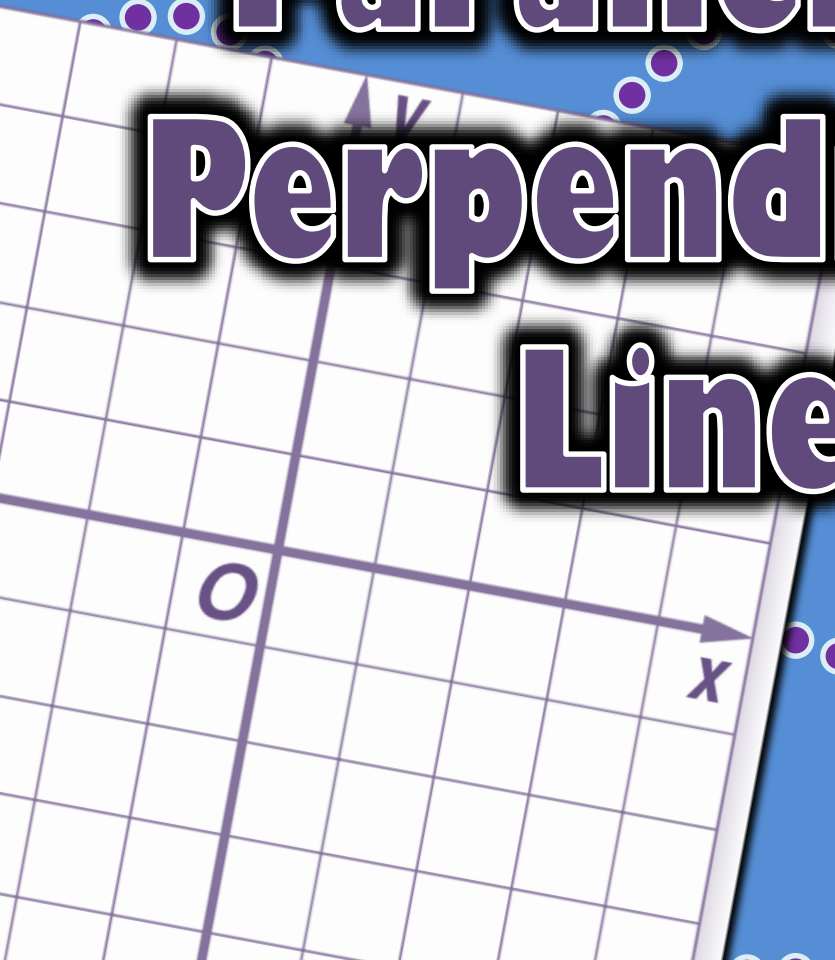


Slopes of Parallel and Perpendicular Lines



Inquiry
Activity

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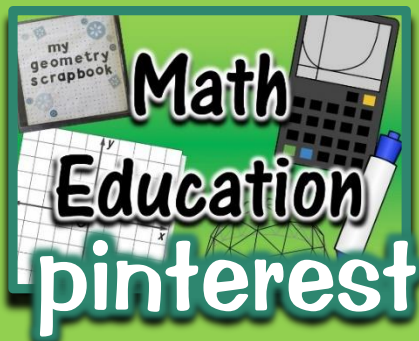
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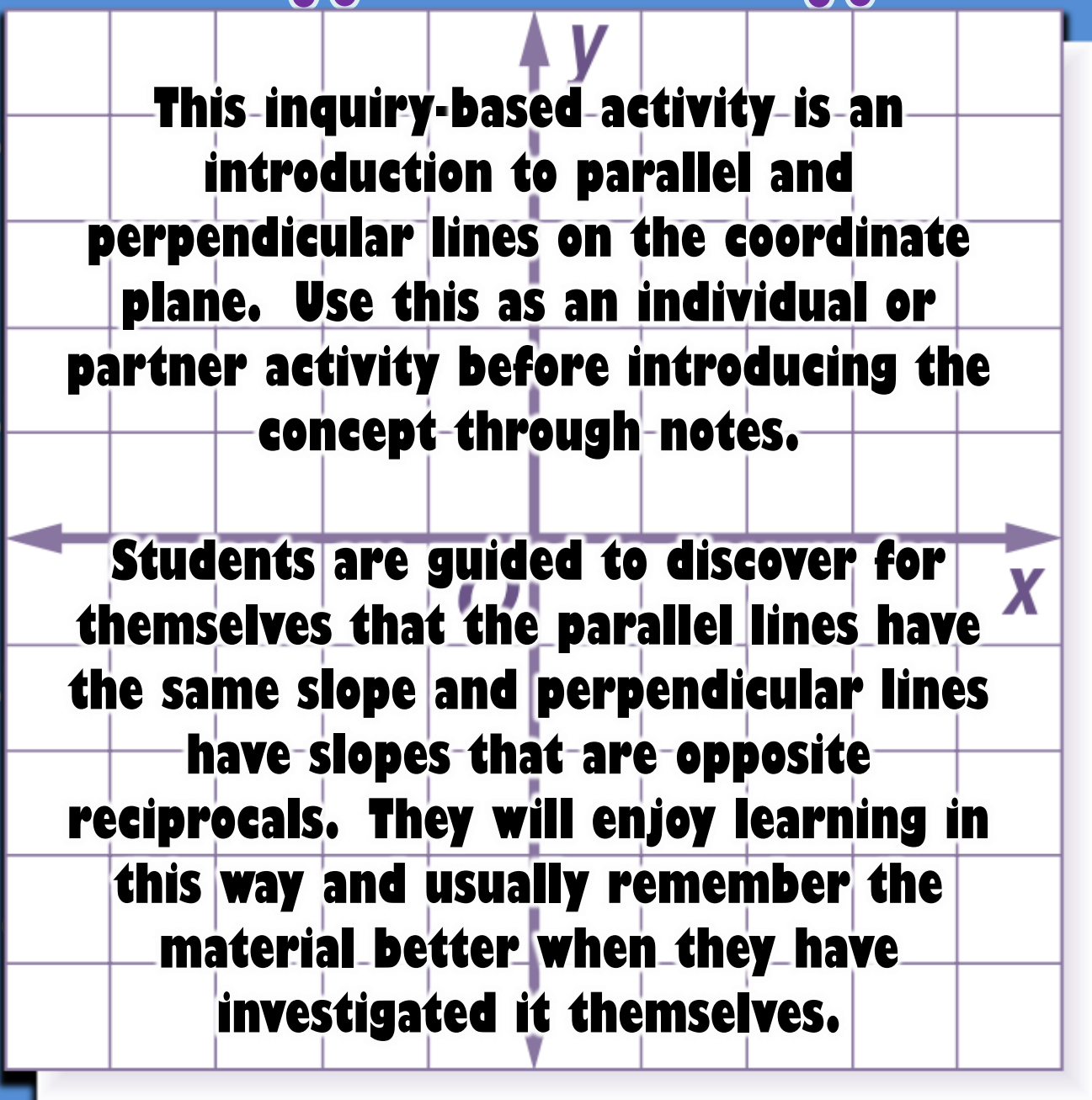
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NOTE TO TEACHER



This inquiry-based activity is an introduction to parallel and perpendicular lines on the coordinate plane. Use this as an individual or partner activity before introducing the concept through notes.

Students are guided to discover for themselves that the parallel lines have the same slope and perpendicular lines have slopes that are opposite reciprocals. They will enjoy learning in this way and usually remember the material better when they have investigated it themselves.

Name: _____

Date: _____ Class: _____

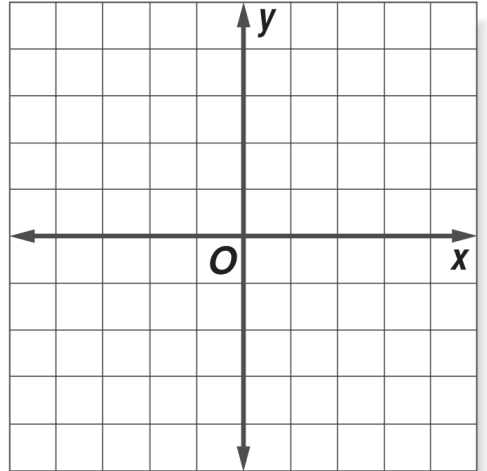
1. Graph both lines on the same coordinate plane.

$$y = 2 - 3x$$

$$y = -3x - 1$$

What do you notice about the two lines?

What do you notice about the equations for the lines?



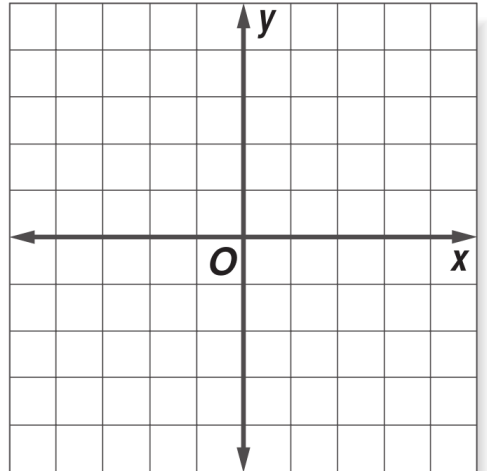
2. Graph both lines on the same coordinate plane.

$$4y = x + 8$$

$$y + 3 = \frac{1}{4}x$$

What do you notice about the two lines?

What do you notice about the equations for the lines?



3. Draw a conclusion that summarizes what you have observed in complete sentences.

4. What is the slope of a line that is parallel to the line $y = -12$?

5. Write the equation for a line that is parallel to the line $y = 5 - \frac{2}{3}x$ and passes through the point $(-3, 0)$.

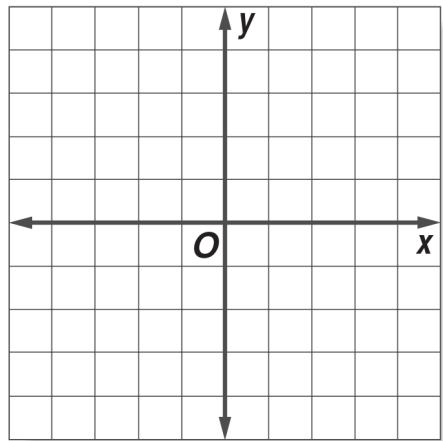
6. Write the equation for a line that is parallel to the line $2x + 3y = 6$ and passes through the point $(0, 4)$



7. Graph both lines on the same coordinate plane.

$$y = \frac{1}{3}x + 3$$

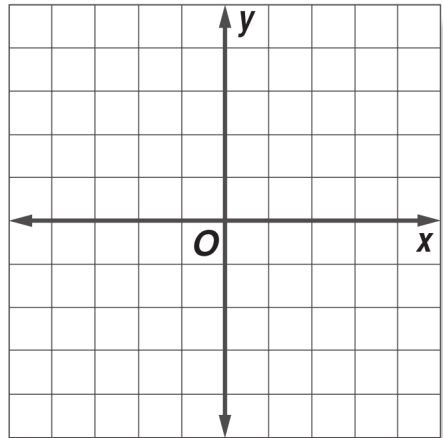
$$y + 1 = -3x$$



8. Graph both lines on the same coordinate plane.

$$2y - 4 = x$$

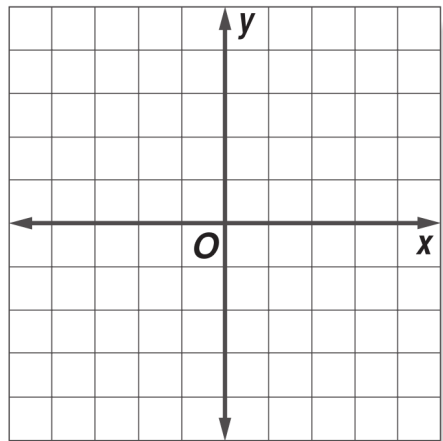
$$Y = 5 - 2x$$



9. Graph both lines on the same coordinate plane.

$$y = \frac{2}{3}x$$

$$-\frac{3}{2}x = y + 2$$



10. What do you notice about the intersection of each pair of lines?

11. Find a pattern in the pairs of slopes for each set of lines.

12. Draw a conclusion that summarizes what you have observed in complete sentences.



13. What is the slope for a line that is perpendicular to the line $y = 4x + 8$?
14. What is the slope for a line that is perpendicular to the line $x = -6$?
15. Write the equation for a line that is perpendicular to the line $8x - 4y = 12$ and passes through the origin.
16. Write an equation for a line that is perpendicular to the line $y = -\frac{1}{3}x$ and passes through the point $(0, -10)$
17. Determine whether each pair of lines is parallel, perpendicular, or neither.

a

$$y = 2x + 6$$
$$y + 1 = -2x$$

b

$$3y - 5x = 9$$
$$y = -\frac{5}{3}x - 12$$

c

$$x + 6 = y$$
$$3y = 3x + 2$$

d

$$\frac{1}{8}y + 17 = 4x$$
$$4x = y + 1$$

e

$$x + y = 0$$
$$y = x + 10$$

f

$$2y - 8 = 5x$$
$$2x - 8 = 5y$$

g

$$y = 6x + 16$$
$$y - 6x = -4$$

h

$$\frac{1}{2}x + 3 = y$$
$$y = -1 - 2x$$



Name: _____

Date: _____ Class: _____

Answer Key

1. Graph both lines on the same coordinate plane.

$$y = 2 - 3x$$

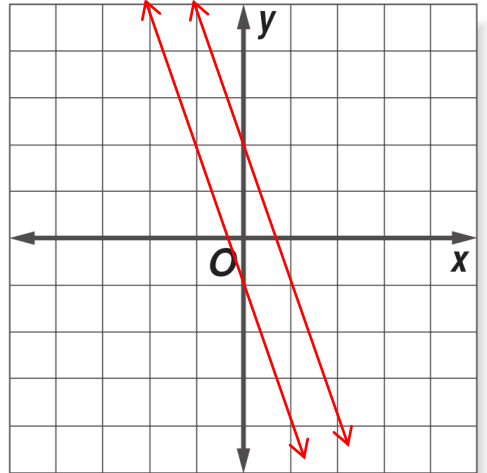
$$y = -3x - 1$$

What do you notice about the two lines?

The lines are parallel.

What do you notice about the equations for the lines?

The slopes are both -3.



2. Graph both lines on the same coordinate plane.

$$4y = x + 8$$

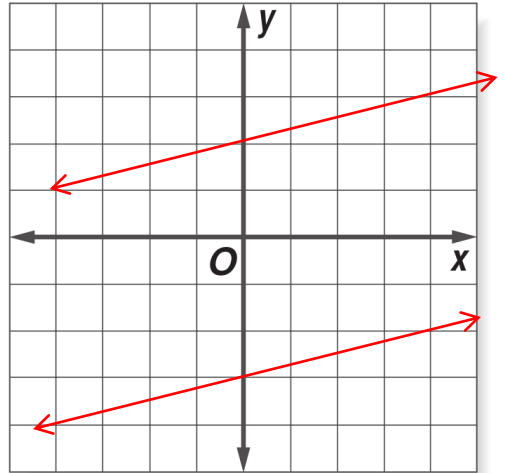
$$y + 3 = \frac{1}{4}x$$

What do you notice about the two lines?

The lines are parallel.

What do you notice about the equations for the lines?

The slopes are both $\frac{1}{4}$.



3. Draw a conclusion that summarizes what you have observed in complete sentences.

Lines that are parallel have the same slope.

4. What is the slope of a line that is parallel to the line $y = -12$?

zero

5. Write the equation for a line that is parallel to the line $y = 5 - \frac{2}{3}x$ and passes through the point $(-3, 0)$.

$$y = -\frac{2}{3}x - 2$$

6. Write the equation for a line that is parallel to the line $2x + 3y = 6$ and passes through the point $(0, 4)$.

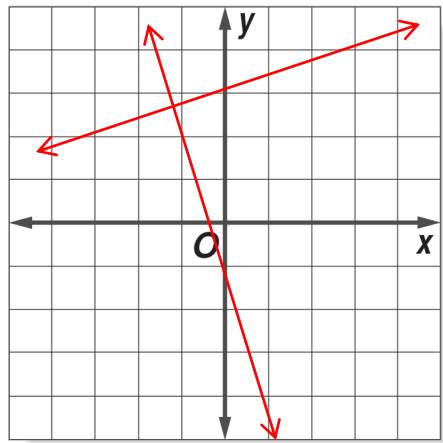
$$y = -\frac{2}{3}x + 4$$



7. Graph both lines on the same coordinate plane.

$$y = \frac{1}{3}x + 3$$

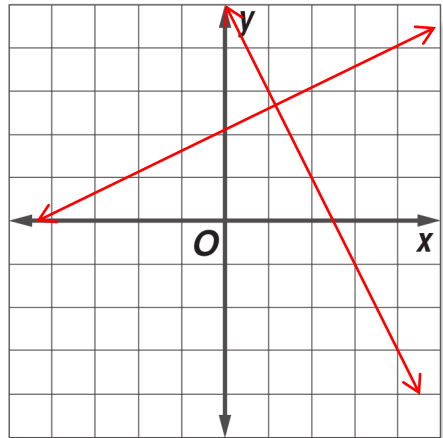
$$y + 1 = -3x$$



8. Graph both lines on the same coordinate plane.

$$2y - 4 = x$$

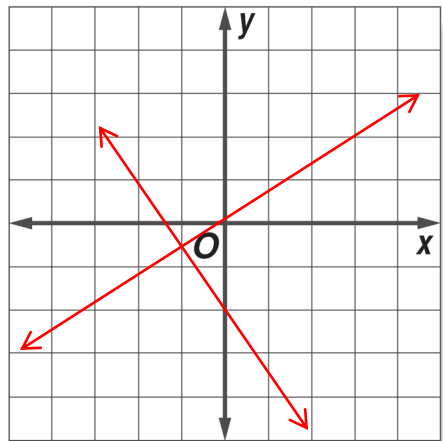
$$y = 5 - 2x$$



9. Graph both lines on the same coordinate plane.

$$y = \frac{2}{3}x$$

$$-\frac{3}{2}x = y + 2$$



10. What do you notice about the intersection of each pair of lines?

The lines are perpendicular.

11. Find a pattern in the pairs of slopes for each set of lines.

The slopes are reciprocals and have opposite signs.

12. Draw a conclusion that summarizes what you have observed in complete sentences.

Lines that are perpendicular have slopes that are opposite reciprocals.



13. What is the slope for a line that is perpendicular to the line $y = 4x + 8$? $-\frac{1}{4}$
14. What is the slope for a line that is perpendicular to the line $x = -6$? zero
15. Write the equation for a line that is perpendicular to the line $8x - 4y = 12$ and passes through the origin.
 $y = -\frac{1}{2}x$
16. Write an equation for a line that is perpendicular to the line $y = -\frac{1}{3}x$ and passes through the point $(0, -10)$
 $y = 3x - 10$
17. Determine whether each pair of lines is parallel, perpendicular, or neither.
- a $y = 2x + 6$ $y + 1 = -2x$ neither
- b $3y - 5x = 9$ $y = -\frac{5}{3}x - 12$ neither
- c $x + 6 = y$ $3y = 3x + 2$ parallel
- d $\frac{1}{8}y + 17 = 4x$ $4x = y + 1$ neither
- e $x + y = 0$ $y = x + 10$ perpendicular
- f $2y - 8 = 5x$ $2x - 8 = 5y$ neither
- g $y = 6x + 16$ $y - 6x = -4$ parallel
- h $\frac{1}{2}x + 3 = y$ $y = -1 - 2x$ perpendicular



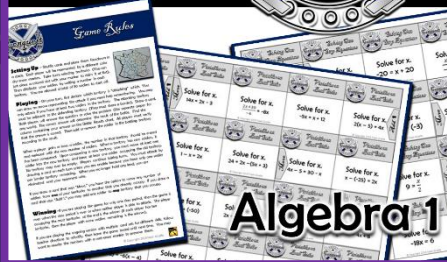
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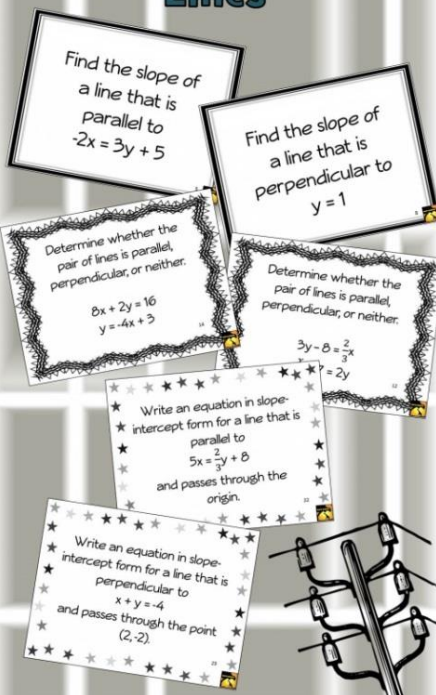
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Algebra 1

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Slopes of Parallel and Perpendicular Lines



Coordinate PLANES

